

# AKAMAI INTELLIGENCE GROUP

INDEPENDENT R&D; LABORATORY -- DISTRIBUTED AUTONOMY GOVERNANCE

## WHO WE ARE

AIG develops governance-first multi-agent AI architectures that remain stable under adversarial conditions -- Byzantine compromise, prompt injection, epistemic drift, and degraded comms -- without central orchestration.

Core thesis: alignment is a property of governance architecture, not isolated models.

## IDENTITY

**Entity:** Hawaii Domestic LLC

**Ownership:** 100% Minority-Owned (Filipino-American)

**SBA 8(a):** Eligible -- application pending

**NAICS:** 541715 / 541512 / 541330

**Patents:** 6 Provisional (AIG-PAT-001 to 006)

**Proposal:** DARPA DICE SBIR Phase I

## CORE CAPABILITIES

|                                                                                                                                                                                            |                                                                                                                                                                        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>CAP-001</b><br><b>Distributed Governance Architecture</b><br>Multi-agent frameworks: bounded authority, compartmentalized cognition, trust-lineage integrity, no central orchestration. | <b>CAP-002</b><br><b>Adversarial Resilience Engineering</b><br>16 attack categories validated. Byzantine tolerance confirmed below 10% node compromise.                |
| <b>CAP-003</b><br><b>Ingestion-Layer IPI Defense</b><br>Infrastructure-level IPI mitigation. 14-category threat taxonomy, cryptographic chain-of-custody.                                  | <b>CAP-004</b><br><b>Stigmergic Coordination Substrates</b><br>Pheromone-analog state buses, 25 signal types, emergent pattern detection.                              |
| <b>CAP-005</b><br><b>Adaptive Network Topology</b><br>Biologically-derived self-optimizing topology. Flow-reinforced conductance. Zero protocol rerouting overhead.                        | <b>CAP-006</b><br><b>Contested Edge Deployment</b><br>Air-gap-capable. 5W sustained compute, zero cloud dependency, local open-source inference.                       |
| <b>CAP-007</b><br><b>Human-AI Governance Interface Design</b><br>HITL gate integrity, trust asymmetry monitoring, 4-level persuasion friction classification.                              | <b>CAP-008</b><br><b>Covert Counterintelligence Architecture</b><br>Out-of-band governance coordination and meta-level identity attack defense. Empirically validated. |

## RESEARCH PORTFOLIO

5 peer-review-ready papers: distributed governance, adversarial resilience, biomimetic coordination topology, covert counterintelligence architecture, and human-AI trust dynamics.

[aig-intel.dev/research.html](https://aig-intel.dev/research.html)

## CONTACT

**Christopher Ramos**

Founder & CEO / Principal Investigator

[research@aig-intel.dev](mailto:research@aig-intel.dev)

[aig-intel.dev](https://aig-intel.dev)